

Exercise 1: Introduction to Modern JavaScript and DOM

Solutions with HTML and JavaScript files

Program	Topic	Files
1(a)	Linking JS file with HTML	index.html, script.js
1(b)	Selecting elements using selectors	selectors.html, selectors.js
1(c)	Event listeners	events.html, events.js
1(d)	Click events on buttons	click.html, click.js
1(e)	Three types of functions	functions.html, functions.js

How to run: Save both files of each program in the **same folder**, then open the .html file in a browser (Chrome/Edge). Press **F12** → **Console** to see console.log() output.

1(a) Linking a JavaScript File with an HTML Page

Question 1(a): Write a JavaScript program to link JavaScript file with the HTML page.

Aim: To link an external JavaScript file to an HTML page using the `<script src="...">` tag.

File 1: index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Linking JavaScript</title>
</head>
<body>
  <h1>Linking JavaScript File with HTML</h1>
  <p id="msg">This text will be changed by JavaScript.</p>

  <!-- Linking the external JavaScript file -->
  <script src="script.js"></script>
</body>
</html>
```

File 2: script.js

```
// This file is linked to index.html
console.log("JavaScript file linked successfully!");

// Change the paragraph text to prove the link works
document.getElementById("msg").innerText =
  "Hello! This message comes from script.js";

alert("Welcome! script.js is connected to index.html");
```

Output

Alert box : Welcome! script.js is connected to index.html
Web page : Hello! This message comes from script.js
Console : JavaScript file linked successfully!

Note: The `<script>` tag is placed just before `</body>` so the HTML elements load first. Alternatively, put it in `<head>` with the **defer** attribute.

1(b) Selecting Elements Using Selectors

Question 1(b): Write a JavaScript program to select the elements in HTML page using selectors.

Aim: To select HTML elements using getElementById, getElementsByClassName, getElementsByTagName, querySelector and querySelectorAll.

File 1: selectors.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>DOM Selectors</title>
</head>
<body>
  <h1 id="heading">DOM Selectors Demo</h1>

  <p class="info">Paragraph 1</p>
  <p class="info">Paragraph 2</p>
  <p class="info">Paragraph 3</p>

  <ul>
    <li>Apple</li>
    <li>Banana</li>
    <li>Mango</li>
  </ul>

  <div id="box">
    <span class="note">First note</span>
    <span class="note">Second note</span>
  </div>

  <script src="selectors.js"></script>
</body>
</html>
```

File 2: selectors.js

```
// 1. Select by ID
let heading = document.getElementById("heading");
heading.style.color = "blue";
console.log("By ID:", heading.innerText);

// 2. Select by Class Name (returns HTMLCollection)
let paras = document.getElementsByClassName("info");
for (let i = 0; i < paras.length; i++) {
  paras[i].style.backgroundColor = "lightyellow";
}
console.log("By Class: found", paras.length, "paragraphs");

// 3. Select by Tag Name
let items = document.getElementsByTagName("li");
console.log("By Tag: found", items.length, "list items");
items[0].style.fontWeight = "bold";

// 4. querySelector - returns FIRST match (CSS selector)
let firstNote = document.querySelector("#box .note");
firstNote.style.color = "red";
console.log("querySelector:", firstNote.innerText);

// 5. querySelectorAll - returns ALL matches (NodeList)
let allNotes = document.querySelectorAll(".note");
allNotes.forEach(function (note) {
  note.style.border = "1px solid green";
});
```

```
});  
console.log("querySelectorAll: found", allNotes.length, "notes");
```

Output (Console)

By ID: DOM Selectors Demo
By Class: found 3 paragraphs
By Tag: found 3 list items
querySelector: First note
querySelectorAll: found 2 notes

On the page: heading turns blue, paragraphs get a yellow background, "Apple" becomes bold, first note turns red, both notes get a green border.

Selector	Returns
getElementById()	Single element
getElementsByClassName()	HTMLCollection (many)
getElementsByTagName()	HTMLCollection (many)
querySelector()	First matching element
querySelectorAll()	NodeList (all matches)

1(c) Implementing Event Listeners

Question 1(c): Write a JavaScript program to implement the event listeners.

Aim: To use `addEventListener()` to handle different events like `mouseover`, `mouseout`, `keyup`, `change` and `double-click`.

File 1: events.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Event Listeners</title>
  <style>
    #box { width: 200px; height: 100px; background: lightblue;
          text-align: center; line-height: 100px; }
  </style>
</head>
<body>
  <h1>Event Listeners Demo</h1>

  <div id="box">Hover over me</div>
  <br>

  <input type="text" id="nameInput" placeholder="Type your name">
  <p id="typed"></p>

  <select id="city">
    <option>Kadapa</option>
    <option>Tirupati</option>
    <option>Hyderabad</option>
  </select>
  <p id="selected"></p>

  <p id="dbl">Double-click this text</p>

  <script src="events.js"></script>
</body>
</html>
```

File 2: events.js

```
let box = document.getElementById("box");

// 1. mouseover event
box.addEventListener("mouseover", function () {
  box.style.background = "orange";
  box.innerText = "Mouse is over!";
});

// 2. mouseout event
box.addEventListener("mouseout", function () {
  box.style.background = "lightblue";
  box.innerText = "Hover over me";
});

// 3. keyup event
let input = document.getElementById("nameInput");
input.addEventListener("keyup", function () {
  document.getElementById("typed").innerText =
    "You typed: " + input.value;
});

// 4. change event
let city = document.getElementById("city");
```

```
city.addEventListener("change", function () {
    document.getElementById("selected").innerText =
        "Selected city: " + city.value;
});

// 5. dblclick event
let dbl = document.getElementById("dbl");
dbl.addEventListener("dblclick", function () {
    dbl.style.color = "purple";
    dbl.innerText = "You double-clicked me!";
});
```

Output

Hover on box	-> box turns orange, shows "Mouse is over!"
Move mouse away	-> box returns to light blue
Type "Vishnu"	-> "You typed: Vishnu"
Choose Tirupati	-> "Selected city: Tirupati"
Double-click text	-> "You double-clicked me!" (purple)

Syntax: `element.addEventListener("eventName", functionToRun);`

1(d) Handling Click Events for Buttons

Question 1(d): Write a JavaScript program to handle the click events for the HTML button elements.

Aim: To handle click events on multiple buttons to change text, change colour, and increase/decrease/reset a counter.

File 1: click.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Button Click Events</title>
</head>
<body>
  <h1>Button Click Events</h1>

  <p id="text">Click a button below</p>
  <button id="greetBtn">Greet</button>
  <button id="colorBtn">Change Colour</button>

  <h2>Counter: <span id="count">0</span></h2>
  <button id="incBtn">+ Increase</button>
  <button id="decBtn">- Decrease</button>
  <button id="resetBtn">Reset</button>

  <script src="click.js"></script>
</body>
</html>
```

File 2: click.js

```
let text = document.getElementById("text");
let countDisplay = document.getElementById("count");
let count = 0;

// Greet button
document.getElementById("greetBtn").addEventListener("click", function () {
  text.innerText = "Hello! Welcome to JavaScript.";
});

// Change colour button
let colors = ["red", "green", "blue", "orange"];
let index = 0;
document.getElementById("colorBtn").addEventListener("click", function () {
  text.style.color = colors[index];
  index = (index + 1) % colors.length;
});

// Increase button
document.getElementById("incBtn").addEventListener("click", function () {
  count++;
  countDisplay.innerText = count;
});

// Decrease button
document.getElementById("decBtn").addEventListener("click", function () {
  count--;
  countDisplay.innerText = count;
});

// Reset button
document.getElementById("resetBtn").addEventListener("click", function () {
  count = 0;
});
```

```
    countDisplay.innerText = count;
    text.innerText = "Click a button below";
    text.style.color = "black";
});
```

Output

Click "Greet"	-> "Hello! Welcome to JavaScript."
Click "Change Colour"	-> text colour cycles red, green, blue, orange
Click "+ Increase" x3	-> Counter: 3
Click "- Decrease" x1	-> Counter: 2
Click "Reset"	-> Counter: 0, text back to original

1(e) Three Types of Functions

Question 1(e): Write a JavaScript program with three types of functions: (i) Function declaration (ii) Function definition (iii) Arrow functions.

Aim: To write and compare a function declaration, a function definition (function expression), and an arrow function.

Note: "Function definition" here means a **function expression** - a function stored in a variable.

File 1: functions.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Types of Functions</title>
</head>
<body>
  <h1>Three Types of Functions</h1>

  <input type="number" id="a" placeholder="Enter first number">
  <input type="number" id="b" placeholder="Enter second number">
  <br><br>
  <button id="declBtn">Add (Declaration)</button>
  <button id="defBtn">Multiply (Definition)</button>
  <button id="arrowBtn">Square of A (Arrow)</button>

  <h3 id="result">Result will appear here</h3>

  <script src="functions.js"></script>
</body>
</html>
```

File 2: functions.js

```
// (i) FUNCTION DECLARATION
// Has a name, is hoisted (can be called before it is written)
function add(x, y) {
  return x + y;
}

// (ii) FUNCTION DEFINITION (Function Expression)
// Function is stored in a variable, NOT hoisted
const multiply = function (x, y) {
  return x * y;
};

// (iii) ARROW FUNCTION
// Short syntax using =>
const square = (x) => x * x;

// ---- Console demo ----
console.log("Declaration   add(5, 3)      =", add(5, 3));
console.log("Definition   multiply(5, 3) =", multiply(5, 3));
console.log("Arrow        square(5)      =", square(5));

// ---- Connect to buttons ----
const getA = () => Number(document.getElementById("a").value);
const getB = () => Number(document.getElementById("b").value);
const show = (msg) => document.getElementById("result").innerText = msg;

document.getElementById("declBtn").addEventListener("click", function () {
  show("Sum = " + add(getA(), getB()));
});
```

```

document.getElementById("defBtn").addEventListener("click", function () {
  show("Product = " + multiply(getA(), getB()));
});

document.getElementById("arrowBtn").addEventListener("click", () => {
  show("Square of A = " + square(getA()));
});

```

Output

Console:

```

Declaration  add(5, 3)      = 8
Definition   multiply(5, 3) = 15
Arrow        square(5)     = 25

```

Web page (A = 6, B = 4):

```

Add (Declaration)    -> Sum = 10
Multiply (Definition) -> Product = 24
Square of A (Arrow)  -> Square of A = 36

```

Type	Syntax	Hoisted?
Declaration	function add(x, y) { ... }	Yes
Definition (Expression)	const multiply = function (x, y) { ... };	No
Arrow	const square = (x) => x * x;	No